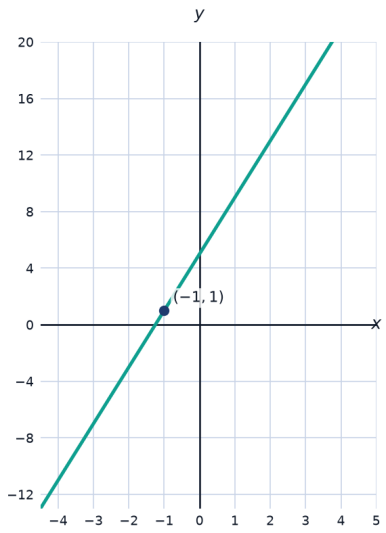
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

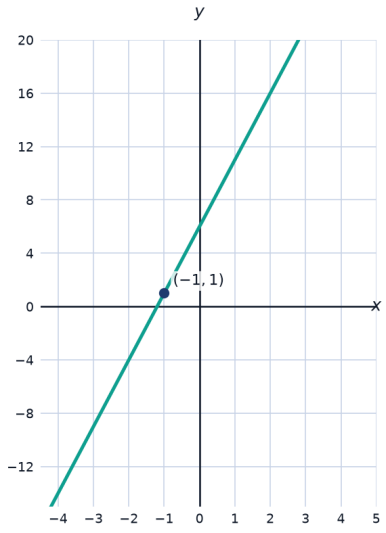
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 6$ in fractions and the number line, evaluate $f(2)$.



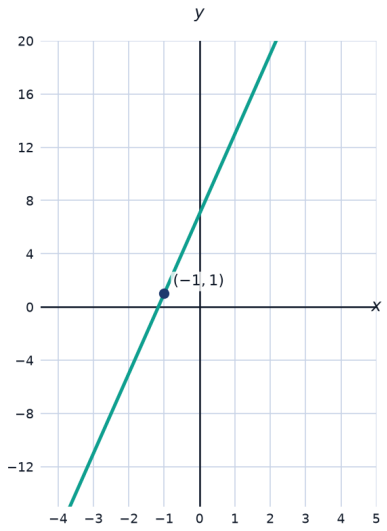
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for fractions and the number line consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 7$ in multiplying and dividing integers, evaluate $f(2)$.

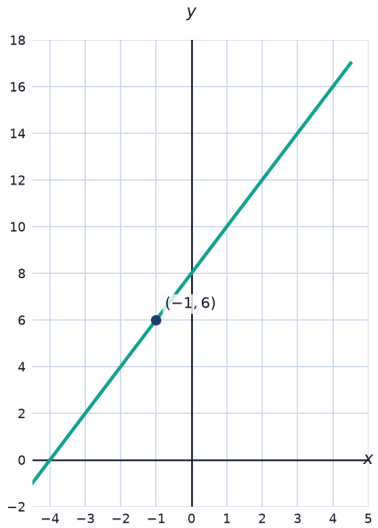


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for multiplying and dividing integers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for mixed operations with integers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

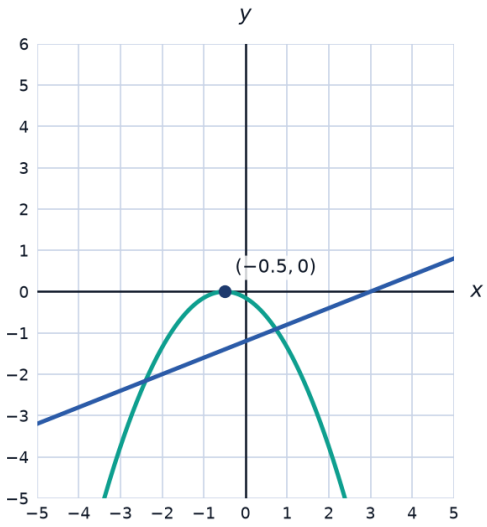
Segment AB has endpoints $A(2,\ -1)$ and $B(6, 7)$. Find the equation of the perpendicular bisector of AB in slope-intercept form.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the equation of the perpendicular bisector of the line segment joining $A(-2, 5)$ and $B(4,\ -3)$. Give your answer in the form $ax + by + c = 0$, where a, b, c are integers.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the coordinates of the point P on directed line segment from $A(-3, 2)$ to $B(7, 12)$ that partitions the segment in the ratio $2 : 3$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the set of values of the constant k for which the equation $kx^2 + 4x + (k - 3) = 0$ has real roots.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circular archery target has three concentric rings of radii $10\ \text{cm}$, $20\ \text{cm}$, and $30\ \text{cm}$. If an arrow hits the target completely at random, what is the exact probability that it lands in the middle ring (between radii $10\ \text{cm}$ and $20\ \text{cm}$)?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A rhombus has diagonals of lengths $16\ \text{cm}$ and $12\ \text{cm}$. Find the perimeter and area of the rhombus.

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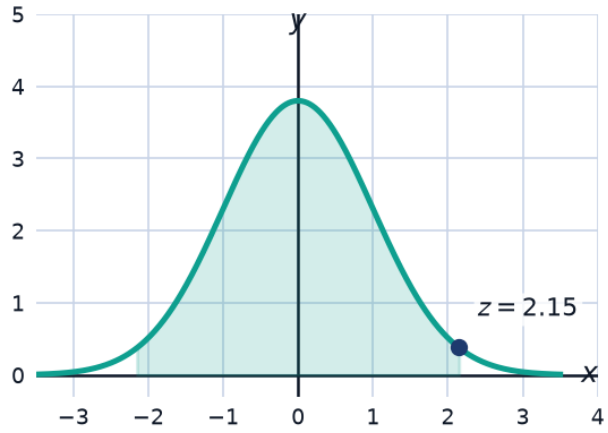
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Sets & Venn Diagrams

From an external point P, two tangents PA and PB are drawn to a circle with radius $r = 5$ cm. The distance from P to the center of the circle O is 13 cm. Find the length of the tangents PA and PB, and find the perimeter of quadrilateral $PAOB$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Algebra & Graphs

A block of mass 8 kg rests in limiting equilibrium on a rough plane inclined at 30° to the horizontal. Taking $g = 9.8$ m/s^2: (a) calculate the normal reaction R, (b) find the coefficient of friction μ between the block and the plane.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Coordinate Geometry

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Mensuration

A train accelerates uniformly from rest to 30 m/s in 20 s, travels at constant speed 30 m/s for 60 s, and then decelerates uniformly to rest in 15 s. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

A block of mass 4 kg is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude 25 N acting at 30° above the horizontal. Taking $g = 9.8$ m/s^2: (a) show that the normal reaction $R = 26.7$ N, (b) find the acceleration of the block.

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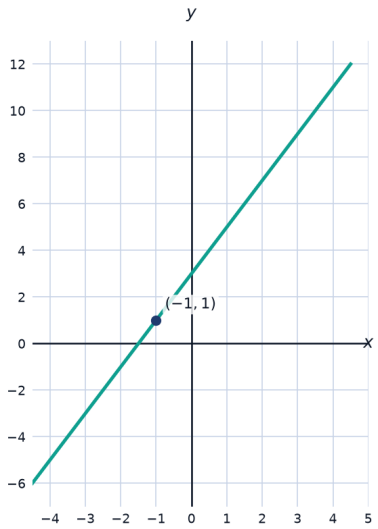
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Transformations & Vectors

Two particles P of mass $3\ \text{kg}$ and Q of mass $5\ \text{kg}$ are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8\ \text{m/s}^2$: (a) find the acceleration a of the system, (b) calculate the tension T in the string, (c) find the force exerted by the string on the pulley.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(2)$.



a. 7

b. 9

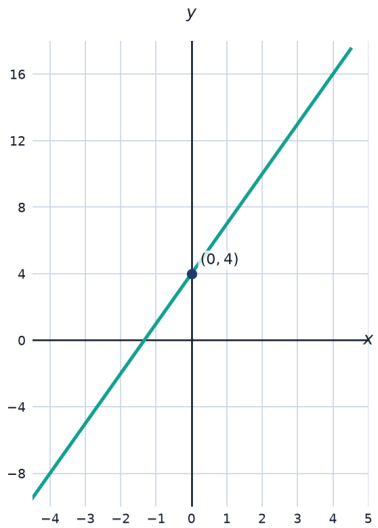
c. 4

d. 11

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in adding and subtracting integers, evaluate $f(2)$.



a. 10

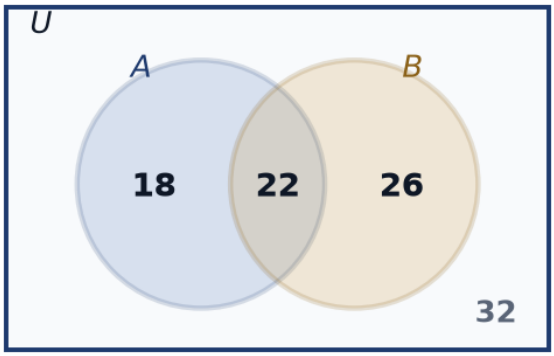
b. 13

c. 6

d. 14

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for integers and the number line consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?



a. 22.5

b. 23.5

c. 24.5

d. 26.5

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for adding and subtracting integers consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A chord of length 24 cm is at a distance of 5 cm from the centre of a circle. What is the radius of the circle?

a. 13 cm

b. 12 cm

c. 14 cm

d. 15 cm

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A tangent TA touches a circle at A. Chord AB makes an angle of 38° with TA. Point C lies on the major arc AB. What is angle ACB?

a. 38°

b. 142°

c. 76°

d. 52°

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A tangent touches a circle at point A. Chord AB makes an angle of 48° with the tangent. By the alternate segment theorem, what is the angle subtended by chord AB in the alternate segment?

a. 48°

b. 132°

c. 42°

d. 96°

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a circle with centre O, point P lies on the circumference. What is the angle between the tangent at P and the radius OP?

a. 45°

b. 60°

c. 90°

d. 180°

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Each interior angle of a regular polygon is 156°. How many sides does the polygon have?

a. 12

b. 15

c. 18

d. 20

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Triangle ABC is inscribed in a circle such that AB is a diameter. If angle BAC = 34°, find the size of angle ABC.

a. 56°

b. 34°

c. 66°

d. 46°

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Points A, B, C, D lie on a circle. Chords AC and BD intersect at X. If angle $\angle ABD = 42^\circ$, what is the size of angle $\angle ACD$?

a. 42°

b. 48°

c. 84°

d. 96°

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Points A, B, and C lie on a circle. AB is a diameter. What is the size of angle $\angle ACB$?

a. 45°

b. 60°

c. 90°

d. 120°

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Points A, B, and C lie on a circle. Angle ABC = 47°. What is the reflex angle AOC subtended at centre O?

a. 266°

b. 94°

c. 188°

d. 274°

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In cyclic quadrilateral PQRS, angle PQR = 104°. What is the size of the opposite angle PSR?

a. 76°

b. 104°

c. 86°

d. 90°

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a regular polygon, the sum of all interior angles is 1440°. How many sides does this polygon have?

a. 10

b. 8

c. 12

d. 9

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a cyclic quadrilateral $ABCD$, angle $A = 75^\circ$. What is the size of the opposite angle C?

a. 105°

b. 75°

c. 127.5°

d. 90°

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The interior angle of a regular polygon is 160°. How many sides does the polygon have?

a. 18

b. 16

c. 20

d. 15

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Mensuration

What is the size of each interior angle of a regular 5-sided polygon?

a. 108°

b. 72°

c. 123°

d. 98°

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

An exterior angle of an isosceles triangle is 110°. If the two non-adjacent opposite angles are equal, what is the measure of each of those two angles?

a. 55°

b. 70°

c. 35°

d. 60°

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Transformations & Vectors

Surface area of a sphere radius 5:

a. 25π

b. 50π

c. 75π

d. 100π

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Probability & Statistics

In a circle, chords AB and CD intersect at point P. If angle ACD = 28°, what is the size of angle ABD?

a. 28°

b. 62°

c. 56°

d. 152°

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Functions

In a regular polygon, each exterior angle is 30°. How many sides does the polygon have?

a. 10

b. 12

c. 14

d. 15

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Number

Triangle DEF is an enlargement of triangle ABC with scale factor 2.5. If the area of triangle ABC is 12 cm², what is the area of triangle DEF?

a. 75 cm²

b. 30 cm²

c. 60 cm²

d. 150 cm²

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Sets & Venn Diagrams

Points A, B, C lie on a circle with centre O. If angle $\angle ABC = 35^\circ$, what is the angle $\angle AOC$ subtended at the centre?

a. 70°

b. 35°

c. 17.5°

d. 145°

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Algebra & Graphs

Two similar triangles have areas in the ratio 16 : 49. What is the ratio of their corresponding perimeters?

a. 4 : 7

b. 16 : 49

c. 2 : 3.5

d. 256 : 2401

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